

Grade 9 ICT – Preparation of Computer Specifications (Notes)

1. Identifying the User

A **user** is a person who uses a computer.

Different users have different tasks and requirements.

User	Task
Programmer	Develops computer programs
Network Administrator	Manages and maintains computer networks
System Analyst	Designs information systems
Software Engineer	Develops software
Computer Application Assistant	Uses office application packages
Web Developer	Develops and maintains websites

Types of Users

1. **System Users** – Manage and maintain software systems.
 2. **End Users** – Use software applications maintained by system users.
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2. Selecting a Computer According to User Requirements

A computer should be selected according to the tasks performed by the user.

Examples of User Requirements

- Accessing the Internet
- Preparing documents
- Editing audio/video
- Playing computer games
- Graphic designing

A computer can be:

1. **Purchased from the market**
 2. **Assembled according to user requirements**
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3. Types of Computers

A. Non-Portable Computers

These computers:

- Need main electricity supply.
- Are large and heavy.
- Are used in fixed locations such as homes, schools, and offices.

Examples:

- Desktop computers
 - Server computers
 - Workstation computers
 - All-in-one computers
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B. Computers for Mobile Use

These computers:

- Use rechargeable batteries.
- Can be carried anywhere.
- Are suitable for travelling.

Examples:

- Laptop computers
 - Notebook computers
 - Tablet computers
 - Smartphones
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4. Computer Peripheral Devices

Peripheral devices are devices connected to a computer to provide input, output, or storage functions.

Types of Peripheral Devices

Input Devices Output Devices Storage Devices

Keyboard	Monitor	Hard Disk Drive
Mouse	Printer	Optical Disk Drive
Microphone	Speakers	Flash Drive
Touch Screen		Magnetic Tape Drive

Note:

A **touch screen** works as both:

- Input device (touching to enter data)
 - Output device (displaying information)
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5. Computer Specifications

Computer specifications describe the features and quality of a computer.

Specifications help users select a computer according to their needs.

Examples of specifications:

- Processor speed
 - Hard disk capacity
 - Memory capacity
 - Monitor size
 - Price
 - Warranty
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6. Technical Specifications of a Computer

6.1 Processor (CPU)

The **processor (CPU)** is the brain of the computer.

Functions:

- Processes data quickly.
- Executes instructions.

- Determines computer performance.
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Processor Speed

Processor speed is the number of instructions executed per second.

Measurement Units:

- Hertz (Hz)
- Megahertz (MHz)
- Gigahertz (GHz)

1000 MHz = 1 GHz

A faster processor:

- Processes more data in less time.
 - Runs software faster.
 - Improves computer performance.
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Processor Manufacturers

Examples:

- Intel
 - AMD
 - Apple
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Types of Processors

Number of CPUs Processor Type

1	Single Core
2	Dual Core
4	Quad Core

Examples:

Type	Examples
Single Core	Pentium I, II, III, IV
Dual Core	Core 2 Duo
Quad Core	Core i3, i5, i7, i9

6.2 Hard Disk

The **hard disk** is the main secondary storage device.

Functions:

- Stores data permanently.
- Stores software and operating systems.

A higher capacity hard disk can store more data.

Units:

- GB (Gigabyte)
- TB (Terabyte)

Examples:

- 500 GB
 - 1 TB
 - 4 TB
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6.3 Monitor Display

The monitor is the main output device.

Users interact with the computer through the monitor.

Monitor Size

- Measured diagonally in inches.
- Larger monitors provide better visibility.

Examples:

- 17 inches

- 19 inches
 - 21 inches
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Monitor Technologies

1. CRT (Cathode Ray Tube)

- Older technology.
- Large and heavy.
- Uses more electricity.

2. LCD (Liquid Crystal Display)

- Smaller and lighter.
- Uses less electricity.

3. LED (Light Emitting Diode)

- Modern technology.
 - Lightweight.
 - Low power consumption.
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6.4 Main Memory (RAM)

Main memory stores data and instructions temporarily while the computer is working.

Features:

- Temporary storage.
- Faster than secondary storage.
- Higher capacity improves performance.

Measurement Units:

- MB (Megabyte)
- GB (Gigabyte)

1024 MB = 1 GB

6.5 Video Graphic Adapter (VGA)

VGA sends output signals to the monitor.

Types of VGA

1. Onboard VGA

- Built into the motherboard.

2. Separate VGA Card

- Added separately.
 - Has its own processor and memory.
 - Useful for gaming and graphic applications.
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Video Ports

VGA Port

- Transfers only images.

HDMI Port

- Transfers both:
 - Video signals
 - Sound signals

Modern computers use:

- DVI
 - HDMI ports
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6.6 Sound Card

A sound card is required for:

- Playing audio
- Recording sounds

Devices connected to sound card:

- Microphone → Input sound
- Speakers → Output sound

Sound Port Colours

Colour	Purpose
Light Green	Audio output (speaker/headphone)
Light Pink	Microphone input
Light Blue	Line input

6.7 Pre-installed Software

An operating system provides communication between the user and hardware.

Software Structure:

User



Application Software



Operating System



Hardware

Types of Operating Systems

Free and Open Source

- Linux
- Ubuntu

Licensed Operating Systems

- Windows 8
- Windows 10

Trial Versions

- Can be used for a limited time.

Before buying a computer, check whether an operating system is already installed.

7. Non-Technical Specifications

7.1 Warranty

Warranty provides protection against defects for a specific period.

Types of Warranty

1. Manufacturer Warranty

- Provided by the manufacturing company.
- Device is repaired or replaced during warranty period.

2. Extended Warranty

- Additional warranty period.
- Requires extra payment.

3. On-site Warranty

- Technicians visit the user's location for repairs.
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7.2 Price

Computer price depends on:

- Processor speed
- Monitor size
- Memory capacity
- Hard disk capacity

A computer should not be selected only based on price.

7.3 After-Sale Services

Services provided after purchasing a computer:

- Technical advice
 - Technical assistance
 - Telephone/email support
 - Website information
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7.4 Ports and Network Connections

Ports allow connection of peripheral devices.

USB Port

USB is the most commonly used port.

Devices connected using USB:

- Printer
 - Keyboard
 - Mouse
 - Scanner
 - External hard disk
 - Digital camera
 - Web camera
 - Pen drive
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Network Port (RJ45)

Used to connect a computer to a wired network.

Wireless Connections

Bluetooth

- Short-range wireless communication.

Wi-Fi

- Wireless network connection.
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8. Important Specifications When Buying a Computer

Technical Specifications

Specification	Examples
CPU Type	Single Core, Dual Core, Quad Core
Processor Speed	2.8 GHz, 3.0 GHz
Memory Capacity	2 GB, 4 GB, 8 GB

Specification	Examples
Hard Disk Capacity	500 GB, 1 TB, 4 TB
Monitor Size	17", 19", 21"
Monitor Type	CRT, LCD, LED
Video Adapter	VGA, DVI, HDMI
Sound Card	Onboard, Separate

Non-Technical Specifications

- Price
- Warranty
- Company reputation
- After-sale service